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Solvent effect on the electrocatalytic nitrogen reduction reaction: a deep potential molecular dynamics simulation with enhanced sampling for the case of the ruthenium single atom catalyst

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The electrocatalytic nitrogen reduction reaction (NRR) is a promising approach for the production of ammonia at ambient temperature and pressure. However, the intricate structure of the solvent–catalyst interface has often led to a significant discrepancy between current computational and experimental studies. To elucidate the explicit hydration structure and complex proton transfer mechanism during electrocatalytic NRR, we employed enhanced sampling molecular dynamics simulations with a neural network potential to incorporate the solvation effect on a nitrogen-doped graphene Ru single-atom catalyst. The present study provides direct observation of how solvent water molecules participate in the NRR process and profoundly impact the NRR dynamics. In particular, a significantly lower free energy barrier (0.59 eV) and a small free energy change (0.15 eV) were obtained for the rate-determining step, *i.e.*, the first hydrogenation step of NRR, compared to the prediction (free energy change of 1.19 eV) without considering the solvent effect. The present prediction is in good agreement with the existing experimental result.

1 Introduction

Today, ammonia (NH_3), a well-known compound with wide applications in agriculture and industries, is receiving special attention for its expanded uses for zero-carbon energy storage and as a fuel.[1–3] The green and mass production of ammonia is being pursued. Electrocatalytic nitrogen reduction reaction (NRR) is a promising method because of its advantages of working under mild reaction conditions, high energy efficiency, and simple process equipment.[4,5]

To develop more efficient electrocatalytic NRR systems, experimental and theoretical researchers are actively working on the reaction mechanism and catalyst design for enhancing the activity, selectivity, stability, and efficiency of NRR.[6–8] With the developments of theories and computer technology, theoretical and computational studies have become increasingly valuable for the NRR catalyst design, which can provide microscopic insights into the catalyst properties and dynamical reaction process, usually unavailable for macroscopic experimental research.[9,10] In this context, recent theoretical advancements have expanded the scope of single-atom catalysts (SACs)[11–13] to dual-metal dimers[14–16] or singe cluster catalyst,[17,18] and beyond traditional carbon supports to more complex 2D-hybrid frameworks, such as functionalized MXenes,[19,20] oxygen modified graphene,[21] BX sheets (X = P, As, Sb),[22] and modified MoS_2.[23,24] These studies have identified a key descriptor for NRR activity: the local electronic structure, which dictates the efficiency of charge transfer from the catalyst to the adsorbed N_2.

Guided by such electronic indicators, theoretical studies can predict catalyst candidates with low limiting potentials to direct experimental synthesis. Nevertheless, discrepancies between the theoretical results and experimental measurements are commonly reported, requiring attention.[25] For instance, the limiting potential U making the electrochemical reaction occur is a very important quantity in practical applications, which is related to the Gibbs free energy barrier (or approximately correlated to the Gibbs free energy change) of the rate-determining elementary step, calculated theoretically at zero electrode potential. However, for many catalysts that have experimentally demonstrated good NRR performance, the theoretical studies predicted a limiting potential much higher than the applied potential in the experiment. For example, the Ru single-atom catalyst supported on nitrogen-doped graphene, despite being experimentally synthesized and demonstrating NRR activity at an applied potential of −0.4 V (*versus* RHE),[26] was later predicted by theoretical studies to require a significantly higher limiting potential of −1.0 V for the reaction to proceed.[26–29] Similar phenomena were observed for other NRR catalysts that have experimentally shown their superiority.[6,30–33]

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There may exist various possible reasons for this discrepancy, such as oversimplification of employed theoretical models, lack of accuracy in electronic structure calculations, unclear mechanisms, *etc.* The oversimplified treatment for the solvent effect to save computational cost is one of the key issues. For example, the computational hydrogen electrode (CHE) model,[34] a classical theory model for approximately obtaining the energy of the solvated proton–electron pair by calculating the energy of the gas-phase hydrogen molecule, is often used to avoid explicit treatment for solvent molecules. Obviously, it oversimplifies the role of the solvent.[35] Recent theoretical research on the carbon dioxide reduction reaction (CO_2RR), the oxygen evolution reaction (OER), and the oxygen reduction reaction (ORR) provided ample evidence that the solvent effect plays an important role in these electrochemical reactions and the interaction with explicit solvent molecules cannot be ignored in simulations.[36–40] The solvent effect has also been noted in the NRR studies. Qian *et al.*[41] constructed a Fe single-atom catalyst model system containing 141 water molecules to systematically investigate the NRR mechanism in water. The results showed that the stability of the reaction intermediates can be significantly affected by interacting with explicit water molecules, leading to a reaction mechanism that is different from that in vacuum. In particular, they found that the inclusion of explicit water molecules in the simulation can even change the assessment of the adsorption spontaneity of the nitrogen molecule on the catalyst.[41]

Although applying the CHE model method to a model system with explicit solvent molecules improves the results, the number of explicit solvent molecules and their equilibrium structures used in the calculations have a direct impact on the final outcome.[42,43] A long-time molecular dynamics (MD) simulation is usually required to obtain the equilibrium solvent environment. For example, in a first-principles MD study of water, Dawson and Gygi reported that it took at least a 15 ps MD simulation to make a 64-molecule water system reach equilibrium, and even then key parameters such as the average number of hydrogen bonds still exhibit significant fluctuations.[44] The challenge is exacerbated in reactive systems. A recent NRR study on the Ru(0001) surface by Liu *et al.*, which used a model with a 16-molecule water layer, found that when nitrogen species were introduced, the surrounding water showed more pronounced structural and energetic fluctuations, and the structures of the reaction intermediates, regardless of whether they were instantaneous configurations sampled from MD simulation or the final structures obtained after geometry optimizations, were highly susceptible to perturbations of the water structure.[45] These findings collectively indicate that static thermodynamic calculations using the CHE model based on a limited number of specific conformations for reaction intermediates solvated in solvent molecules, may fail to provide an accurate description of the reaction mechanism and dynamics.

A simplified kinetic analysis based on the transition state located for each elementary step and the corresponding energy barrier using the nudged elastic band (NEB) method[46,47] also faces similar challenges. For complex catalytic reactions, in a given reaction state, multiple configurations can collectively contribute to the free energy and affect the activation barriers, especially when the solvent molecules participate in the reaction directly or indirectly. For example, in NRR with water as the proton source, each hydrogenation step may involve concerted proton transfer (PT) between the reaction intermediate and multiple solvent molecules *via* the Grotthuss mechanism,[48,49] so it may not be suitable to use the NEB method to find the dividing surface of the reaction for such complicated cases, as it requires well-defined reaction coordinates to approximate the reaction path.

With the rapid advancement of computational capabilities, dynamic methods, such as those calculating the potential of mean force (PMF) along a defined reaction path by the slow-growth method, have made it feasible to address this challenge.[25,38,41,50,51] Furthermore, factors such as nuclear quantum effects, cation effects, and constant potential treatment *via* a grand canonical approach can also be taken into account in these dynamic simulations.[39,52–56] By extracting the average forces along a pre-defined reaction coordinate from the constrained molecular dynamics simulations, which correspond to the derivatives of the free energy, the free energy surface of the reaction can be reconstructed. However, constrained molecular dynamics often lack a driving force for sampling degrees of freedom beyond the reaction coordinate, such as the rearrangement of the solvent water layer. This means that a longer simulation time is needed to equilibrate the system in these degrees of freedom, and thus the average force along the reaction coordinates may be difficult to fully converge. In addition, it struggles to acquire multidimensional free energy surfaces, thereby limiting the depth of analysis.

In this work, we systematically investigated the solvent effect of NRR by molecular dynamics simulation based on the on-the-fly probability enhanced sampling (OPES)[57,58] using the Voronoi collective variables (CVs).[59–62] We defined the protonation coordinate CV, s_p, and the charge–charge distance CV, s_d, to distinguish various states during hydrogenation in the NRR process which can help to take into account complex multiple proton transfer processes among the reaction system and the water network. Although Voronoi CVs-based OPES accelerates the sampling of reactions, the computational cost of AIMD simulations for an explicit solvent model system with a large number of water molecules, is still high at the commonly used density functional theory (DFT) level.

Recent advancements in machine learning (ML) potentials have significantly improved interfacial simulations. Approaches like the Gaussian Approximation Potentials (GAP) and Sparse Gaussian Process Regression (SGPR) offer high data efficiency,[63,64] the Neuroevolution Potential (NEP) model demonstrates an extremely fast simulation rate,[65,66] while Message-Passing Neural Networks (MPNNs) including recent constant-potential implementations by Chen *et al.*[67] and Wang *et al.*[68] show great promise for capturing long-range interactions. Considering that the OPES method necessitates relatively long and stable simulations, the Deep Potential (DP) method was selected among these options for its established ability to balance *ab initio* accuracy and the efficiency required for simulating complex reactive chemistry,[69,70] as well as its

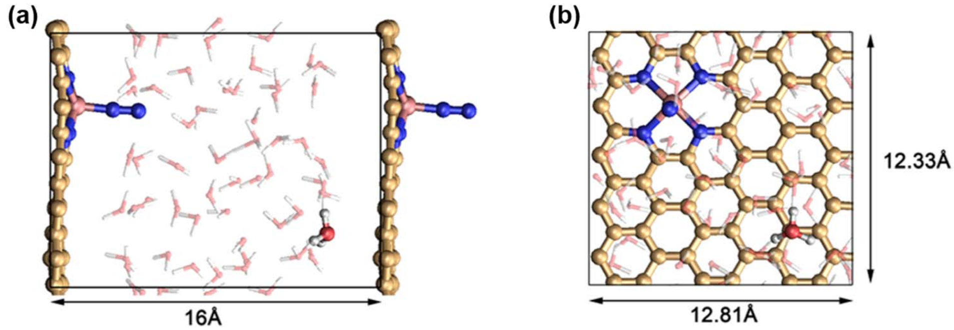

Fig. 1 Model system. (a) Side view; (b) top view. Color legend of atoms: Ru, purple; O, red; N, blue; C, yellow; H, white.

superior stability and robustness. Furthermore, DP has proven effective for various reactions in aqueous solutions and at interfaces.[61,71–78]

In particular, we selected the Ru single-atom catalyst supported on nitrogen-doped graphene (Ru–N_4–C) as the target catalyst (Fig. 1) for three reasons: (1) there were abundant studies on it, and it had been experimentally synthesized and shown to exhibit a superior catalytic performance on NRR; (2) as mentioned earlier, there is a large discrepancy between the experimental observation (with an applied potential of approximately −0.4 V or even lower (relative to RHE)[26,79]) and theoretical prediction (around −1.0 V (ref. 27–29, and 80)); (3) Ru–N_4–C catalyst has a well-defined reaction site, and the basic consensus on the reaction mechanism has been reached. That is, on the Ru–N_4–C catalyst, nitrogen prefers to be adsorbed at the Ru site in an end-on mode, and after N_2 adsorption, the adsorbed N_2 (*N_2) will go through six hydrogenation steps to produce NH_3, in which the first hydrogenation step, *N_2 + H^+ + e^- → *NNH is the rate-determining step of NRR.[27–29,79] Therefore, the present deep potential molecular dynamics (DPMD) study focused only on the key steps of the NRR process, namely, the competitive adsorption of nitrogen gas and hydrogen atoms on the active reactive site and the rate-determining step, to analyze the solvent effects.

2 Computational details

2.1 DFT calculations

All periodic DFT calculations, including the CHE model calculations and single-point calculations for building DP data sets, were performed with the CP2K package.[81] The catalytic system was modeled based on a 5 × 3 × 1 supercell of N-doped graphene with one Ru atom anchored in the center of four N atoms,[27,79,82] with dimensions of 12.33 × 12.81 × 16.00 Å^3, in which one N_2, 53 H_2O molecules and one H atom were added between the periodic catalyst surfaces to mimic the NRR in a real water environment with an approximate density of 1 g cm^{-3}.[41,78,83,84] The side and top views of the model system are shown in Fig. 1. The Perdew–Burke–Ernzerhof (PBE) exchange–correlation functional[85] supplemented with Grimme's D3 correction for dispersion[86,87] (PBE + D3) was used because it was effective and had been widely used in describing the electrical, structural, and dynamic properties of single-atom catalysis in water.[36,41,51] Wave functions with molecule-optimized (MOLOPT) double-ζ Gaussian basis sets[88] were employed, with a cutoff energy of 800 Rydberg. Electrons were modeled by scalar relativistic (SR) norm-conserving Goedecker, Teter, and Hutter (GTH) pseudopotentials with 16, 4, 5, 6, 1 valence electrons for Ru, C, N, O, H, respectively.[89–91] The *k*-point sampling was restricted to the Γ point. The energy convergence threshold was set to 3.0 × 10^{-7} hartree in self-consistent field calculations, and for structural optimization, the convergence criterion for the maximum force component was set to 1.0 × 10^{-4} bohr^{-1} hartree. For post-processing analyses, the CM5 charges[92] were calculated to quantify charge distribution using Multiwfn.[93,94]

2.2 DP model training

The DP model was trained using the DeePMD-kit package,[69,70] based on the reference energies and forces calculated by the PBE+D3 method. The primary data set was generated by DP generator (DP-GEN),[95] which iteratively executes three steps: DP potential training using DeePMD-kit, DPMD simulation with enhanced sampling using LAMMPS,[96] and PBE+D3 single point calculations using CP2K. Enhanced sampling in dynamic processes was implemented by the PLUMED[97] software. The DPGEN active learning process iteratively explored configurational space without any geometric constraints, and both end-on and side-on adsorption modes of N_2 on the catalyst had been generated and included in the training database. A total of 20 266 structures obtained through the DP-GEN workflow were used for the final DP model training. More computational details related to the training procedure are given in the SI.

2.3 DPMD simulations with OPES

After acquiring the final DP model, DPMD simulations with the OPES method were carried out in the canonical (NVT) ensemble using a time step of 0.5 fs. A 10 ns simulation at 300 K was conducted for each of N_2 adsorption, proton adsorption, and the initial hydrogenation processes. Convergence was verified by monitoring the time evolution of free energy profiles (shown in Fig. S3–S6 in SI), which indicated that the 10 ns duration was sufficient. This simulation timescale exceeds those in comparable studies.[40,78] The constant temperature was enforced using the Nosé–Hoover thermostat.[98,99] We used mainly two types of CV to describe reactions in the water environment.[59,60,62] The first is the protonation state CV, s_p, which indicates the protonation status of the reaction sites and is therefore used to determine the progress of the protonation reaction. In particular, $s_p = -1$ denotes that the reaction site (N or Ru) is unprotonated, and $s_p = 0$ signifies that one hydrogen atom is adsorbed. The second variable is the charge–charge distance CV, s_d, representing the sum of the distances between ions within the system (for example, H_3O^+ and OH^-). The variable s_d is mainly used to assist in the analysis of the ion distribution during the reaction. We also considered two additional CVs: the shortest distance between the oxygen atoms of water molecules and the Ru site, dO_{min}, and the average distance between the N atoms of the nitrogen molecule and the Ru site, dN_{mean}, which were used to describe the adsorption states of water molecules and nitrogen molecules on the catalyst. The details and implementation of all these CVs can be found in the SI.

2.4 CHE model

For comparison, we also employed the CHE model[34] to calculate the free energy change (ΔG_{CHE}) for the system without including any explicit solvent molecule, as shown in the following equation

$$\Delta G_{CHE} = \Delta E + \Delta E_{ZPE} - T\Delta S + eU + G_{pH},$$

where ΔE is the reaction energy directly obtained from DFT calculations; ΔE_{ZPE} and ΔS are the changes of zero-point energy and entropy after the reaction, respectively. The zero-point energies and the entropies of the adsorbates are calculated using the vibrational frequencies of the optimized structures. When the free energy of the surface adsorption structure is corrected, frequencies below 50 cm^{-1} are treated as 50 cm^{-1}.[100] The electrochemical free energy contribution related to an applied potential U is approximately estimated by eU. ΔG_{pH} is the free energy correction of pH, which can be calculated by

$$\Delta G_{pH} = k_B T \times \text{pH} \times \ln 10,$$

where k_B is the Boltzmann constant.

In addition, some auxiliary software was used: ASE for thermodynamic correction,[101] VESTA[102] and VMD[103] for visualization, and MEPSA for computing the minimum free energy pathway.[104]

3 Results and discussion

3.1 Validation of the DP model

First, we validated the accuracy of the DP model in describing the water environment. We performed a short trajectory simulation for the current system with the same initial condition using AIMD and DPMD respectively, and then compared their results. The radial distribution functions (RDF) $g(r)$ of the selected atomic pairs in the water environment obtained by the AIMD and DPMD are plotted in Fig. 2a. We observed good agreement of DPMD with AIMD for this short trajectory, indicating that the DP model approximately approaches the accuracy of PBE+D3 in describing the water structures.

In addition, we constructed a test data set by sampling configurations from three independent enhanced-sampling DPMD simulations to further validate the accuracy of the DP model. These three simulations simulated the processes of the adsorption of nitrogen molecules on the catalyst, the adsorption of hydrogen atoms on the catalyst, and the first protonation process of *NN, respectively. As shown in Fig. 2b, the root mean square errors (RMSEs) of the DP model compared to the PBE+D3 method for the test data set are 0.88 meV per atom for energy and 60 meV Å^{-1} for force, respectively. This shows that the DP model can achieve satisfactory accuracy when simulating our target reactions.

3.2 N_2 and proton adsorption

Herein, N_2 adsorption and proton adsorption reactions on the catalyst are also referred to as R1 and R2 for convenience, respectively. In the enhanced-sampling DPMD simulation for the reaction R1, the CVs were chosen as the average distance of N in N_2 to Ru (denoted as dN_{mean}) and the minimum distance of O in H_2O to Ru (denoted as dO_{min}) in order to not only capture the event of N_2 adsorption but also depict the adsorption of a water molecule at the same Ru site. We investigated the adsorption of water here because it had been previously reported[84] that water molecules could adsorb on the backside of a single-atom catalyst to potentially affect the catalytic process due to the periodic system setup in the calculations. As shown in Fig. 3a, which plots the two-dimensional (2D) free energy surface (FES) of R1 as a function of dO_{min} and dN_{mean}, four wells of FES (marked with a, b, c, and d in the figure) are located, corresponding to the stable states of before and after N_2 or/and H_2O adsorption on the Ru–N_4–C catalyst, respectively. The corresponding structures are displayed in Fig. 3b and labeled as [R1-a/b/c/d].

In accordance with the values of dO_{min} and dN_{mean} of these wells, well a is identified as a H_2O-only adsorption structure ([R1-a]), well b is a co-adsorption structure of N_2 and H_2O ([R1-b]), well c is a N_2-only adsorption structure ([R1-c]), and well d is a structure without adsorption of either species ([R1-d]). [R1-b] and [R1-c] show that N_2 prefers an end-on adsorption mode on the Ru–N_4–C catalyst when explicit water molecules are considered, which is consistent with the preferred end-on N_2 adsorption revealed by the structural optimization in vacuum. It can be seen that the adsorption of water molecules on the backside is also observed in the current simulation, but it is very weak and hardly affects the N_2 adsorption either in energy or geometry. Therefore, in the following investigation, we will not consider the adsorption of water molecules on the catalyst.

For the reaction R2, Voronoi CVs s_p and s_d are used in the enhanced-sampling DPMD simulations, in which the protonation coordinate s_p helps to identify the proton adsorption process on the Ru site, and the ion-distance coordinate s_d is a measurement of the distance between H_3O^+ and OH^- ions in the solution, which indicates whether the water self-ionization occurs during the proton adsorption process. The two-dimensional FES of R2 along s_p and s_d coordinates (Fig. 4a) locates three obvious configurations (a, b, and c), which are

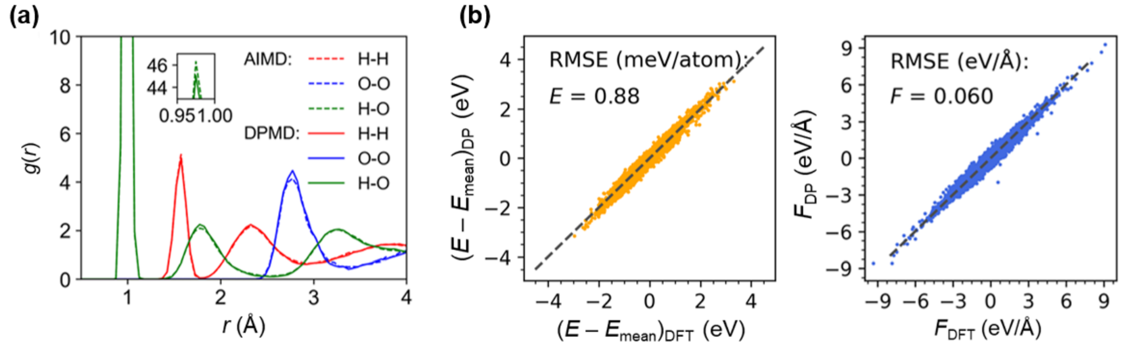

Fig. 2 DP model validation. (a) Radial pair distribution function $g(r)$ of the selected atom pairs. (b) Error distribution of the DP model. The DFT energies and forces are calculated by the PBE+D3 method; E_{mean} refers to the average value of energy.

identified and displayed in Fig. 4b as [R2-a], [R2-b], and [R2-c], respectively. [R2-a] is the configuration before the proton adsorption in which the hydronium (H_3O^+) is in the aqueous solution with $s_p = -1$. Both [R2-b] and [R2-c] are the configurations after the proton adsorption on the Ru atom whose s_p values are both 0, but compared to [R2-b], [R2-c] has a larger s_d, indicating the co-existence of H_3O^+ and OH^- ion pairs in the system. [R2-c] is much less stable and can easily transform to more stable [R2-b] by crossing a very low energy barrier (0.1 eV).

The 2D FES of R2 shows that the only favorable proton adsorption pathway is directly from [R2-a] to [R2-b], that is, H_3O^+ first diffuses to the region close to the Ru active site and then transfers the proton to the catalyst to form the H-adsorbed (*H) structure. Fig. 4c displays several representative structures along this optimal path, which show the dynamic details of the proton adsorption process and how the hydrogen bond structure of the reactant hydronium ion changes during the reaction process.

To more easily compare the energetics of N_2 and proton adsorption on the Ru–N_4–C catalyst, we projected the 2D FES in Fig. 3a and 4a along the CV dN_{mean} and the CV s_p, respectively. The one-dimensional (1D) FES profile along the CV dN_{mean} shown in Fig. 3b indicates that N_2 adsorption involves a notable free energy barrier of approximately 0.3 eV, and the free energy difference before and after adsorption is around −0.7 eV, smaller than the CHE result (−1.1 eV) in absolute values. The 1D FES profile (Fig. 4b) of R2 along the CV s_p shows that compared to the N_2 adsorption, the proton adsorption on the Ru–N_4–C catalyst needs to conquer a similar free energy barrier of about 0.3 eV, but with a much weaker reaction free energy of −0.3 eV. The proton adsorption free energy again deviates remarkably from the prediction of the CHE model (−0.8 eV). By comparing the predicted N_2 and proton adsorption free energies obtained by the DPMD (−0.7 and −0.3 eV) with those by the CHE model (−1.1 eV and −0.8 eV), we conclude that the solvent effect significantly weakens the adsorption of the N_2 and proton on the Ru–N_4–C. Based on the current DPMD results, the N_2 adsorption is more competitive compared to the proton adsorption because they have similar free energy barriers but the former is more exergonic. However, we note that the nuclear quantum effect (NQE) can accelerate proton adsorption dynamics by lowering the free energy barrier[54] but can barely affect the N_2 adsorption at electrocatalytic temperatures, which was not considered in this work. Nevertheless, from

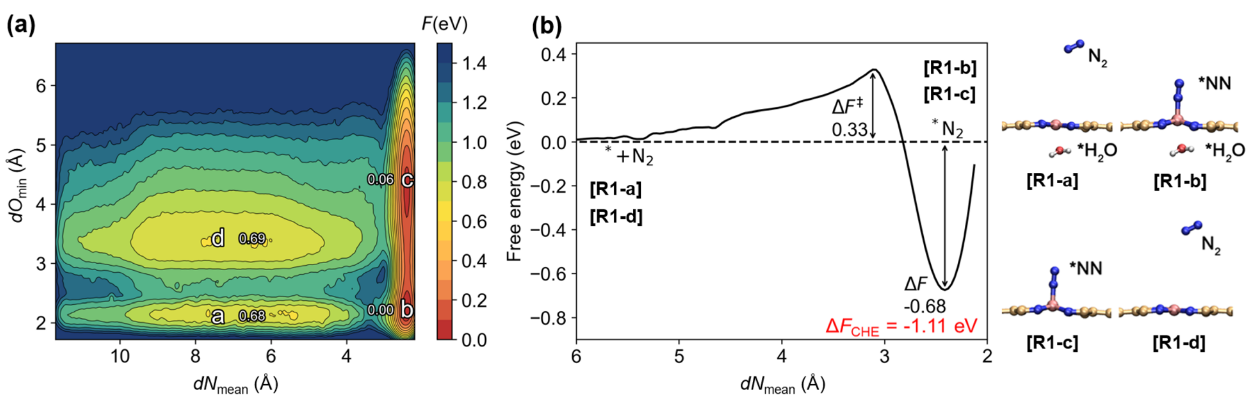

Fig. 3 Free energy surfaces of N_2 adsorption (R1: * + N_2 → *N_2) on Ru–N_4–C catalyst in water. (a) Two-dimensional FES of R1 as a function of dO_{min} and dN_{mean}. (b) One-dimensional projection of the FES of R1 along the dN_{mean}, in which $\Delta F^{\ddagger}$ denotes the free energy barrier, and ΔF denotes the reaction free energy. For comparison, the CHE results in vacuum are marked with red font. Key structures located in R1 are illustrated in the right, with other surrounding water molecules hidden. In the labels, the asterisk * denotes the catalyst, for instance, *N_2/H_2O denotes that the N_2/H_2O is adsorbed on the Ru site of the catalyst.

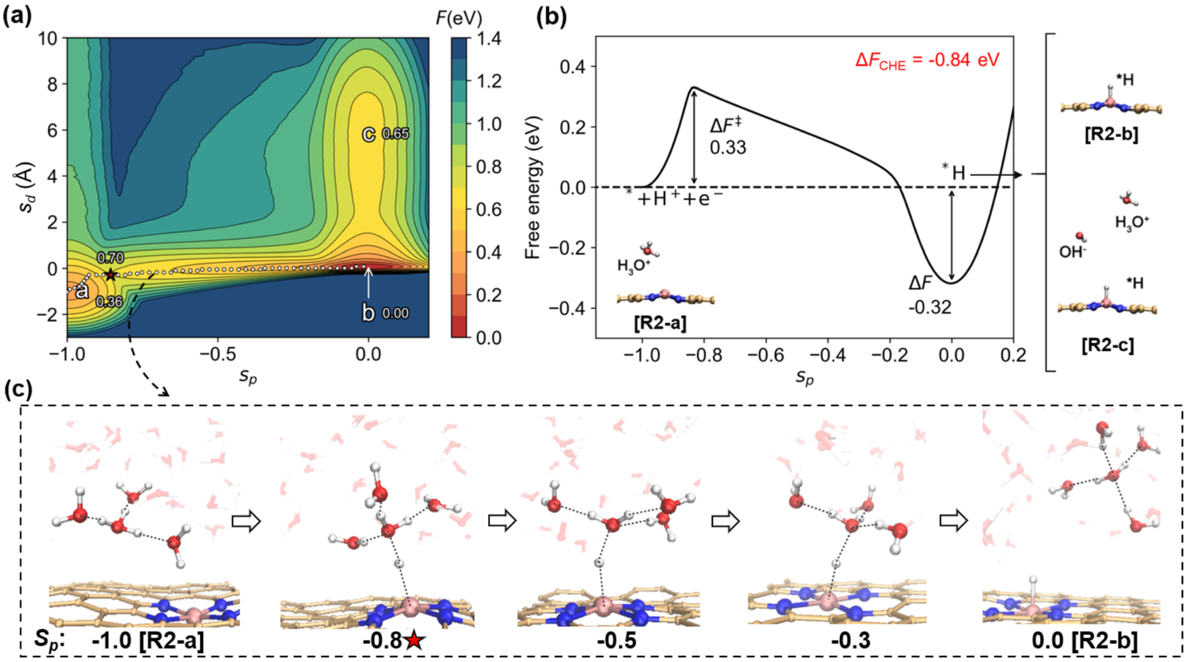

Fig. 4 Free energy surfaces of hydrogen adsorption (R2: * + H^+ + e^- → *H) on Ru–N_4–C catalyst in water. (a) Two-dimensional FES of R2 as a function of the protonation coordinate s_p and the charge–charge distance s_d, where the minimum free energy path (MFEP) along the reaction coordinate is represented by the white dot line. The specific free energies of some selected points along MFEP are given, and the location of the maximum of the MFEP is marked with a star symbol. (b) One-dimensional projection of the FES of R2 along the CV s_p. Key structures located in R2 are marked, with other surrounding water molecules hidden. (c) Some representative configurations along the MFEP, which show the dynamic process of R2.

a thermodynamic point of view, we can still conclude that the N_2 adsorption is more favorable on the Ru–N_4–C catalyst than the proton adsorption.

3.3 First hydrogenation step of NRR

Next, we focus on the effect of explicit water molecules on the first hydrogenation step of NRR (R3), which has been identified as the rate-determining step in previous studies.[27–29,79] We simulated the R3 with the similar enhanced-sampling setting to that in the simulation for R2, *i.e.*, using Voronoi CVs s_p and s_d, and the only difference was to choose the distal N atom of the adsorbed N_2 (*N_2) as the protonation site instead of the Ru atom in calculating the protonation coordinate s_p. The two-dimensional FES of R3 along the CVs s_p and s_d is plotted in Fig. 5a, which successfully locates the stable states before and after the hydrogenation reaction R3 (marked by a, b and c in the figure). The corresponding configurations, [R3-a], [R3-b], and [R3-c], are displayed in Fig. 5b. [R3-b] and [R3-c] are the configurations after hydrogenation, and in the latter, an ion pair of hydronium and hydroxide is observed in the water.

The [R3-a] that represents the reactant state before hydrogenation is characterized by the presence of a hydronium ion in the system, but the exact position of this ion relative to the *NN in its optimal configuration can not be identified from the present 2D FES. Therefore, we further applied a new Voronoi CV s_{d2} (details refer to SI) to understand it, which characterizes the separation of the hydronium ion and the adsorbed N_2. We find that the system is slightly more stable when the hydronium ion is fully solvated in water, and thus the approach process of the hydronium ion in water towards the *N_2 is slightly endergonic with a small free energy change of 0.1 eV, based on the one-dimensional free energy curve along the s_{d2} shown in Fig. S1.

Combining the results of Fig. 5a and S1, we reveal two reaction pathways of R3. The primary reaction pathway (R3-P1) is from [R3-a] to [R3-b]. That is, in this pathway, the fully solvated hydronium ion in water first approaches the *N_2 and then directly transfers the proton of the nearby hydronium ion to *N_2 to achieve the first hydrogenation step. The detailed dynamic process along the R3-P1 path is illustrated in Fig. 5c, including the structure change of the hydrogen bonds surrounding the reaction site. The other path (R3-P2) is [R3-a] → [R3-c] → [R3-b], in which *N_2 first abstracts the proton of the nearby water molecule to produce *NNH and hydroxide ion ([R3-c]), and then [R3-c] converts to more stable [R3-b]. In the current acidic water environment, the R3-P2 pathway that takes water molecules as the proton source is slower in dynamics due to a higher free-energy barrier of around 1.1 eV than the R3-P1 pathway, which is a direct proton transfer pathway with the hydronium ion as the proton source and has a free-energy barrier of around 0.6 eV.

We further projected the 2D free energy surface in Fig. 5a along the protonation coordinate s_p, and the obtained one-dimensional free energy diagram in Fig. 5b shows that the first protonation step of NRR on the Ru–N_4–C catalyst is only slightly endergonic with a reaction free energy of 0.1 eV based on the current simulations with explicit solvent molecules. This

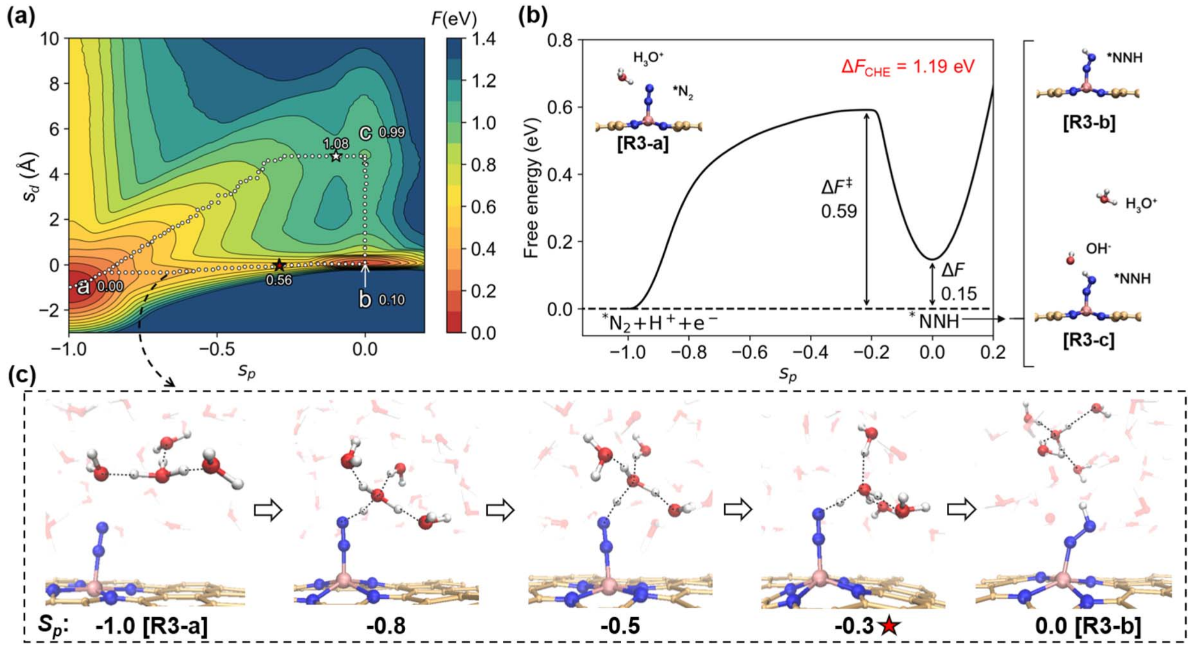

Fig. 5 Free energy surfaces of the first hydrogenation step (R3: $^*N_2 + H^+ + e^- \rightarrow {}^*NNH$). (a) Two-dimensional FES as a function of the protonation coordinate s_p and the charge–charge distance s_d, where minimum free energy paths along the two pathways of R3 are represented by the white dot lines. The specific free energies of some selected points along the MFEPs are given, and the maxima of the MFEPs are marked with a star symbol. (b) One-dimensional projection of the FES along the CV s_p. Key structures located in R3 are marked, with other surrounding water molecules hidden. (c) Some representative configurations along the R3-P1 path, which show the dynamic details along the reaction path.

is significantly lower than the estimated free energy (1.2 eV) of reaction using the CHE model without explicitly considering the water environment. The free energy barrier of R3, which is the rate-determining step of NRR, is 0.6 eV based on the 1D FES in Fig. 5b, and thus, the theoretical applied potential is approximated to be −0.6 V, in a good agreement with the experimental applied potential (around −0.4 V).[26,79]

The results underscore the impact of the explicit aqueous environment and ionic species on the NRR. Qualitatively scrutinizing this influence offers guidance for catalyst design. Studies with the conventional CHE model typically identify the first protonation step as the rate determining step, implying that a good NRR catalyst should kinetically promote the *NN activation and thermodynamically strengthen the *NNH stabilization. Our explicit solvation simulations reveal a more nuanced picture.

As shown in Fig. 6, the charge density difference (CDD) analysis and the CM5 charge calculations reveal a stronger interaction of the water environment with *NNH than *NN *via* hydrogen bonds, which well explains the significantly reduced free energy change of the first hydrogenation step in water compared to that in vacuum. Although this theoretically implies a potential shift in the rate-determining step or reaction mechanism as suggested by previous literature,[45,105,106] current data limit our interpretation to the solvent's heterogeneous stabilizing effect.

Previous literature has proposed that *NN activation relies on electron feedback from the catalyst into the NN anti-bonding orbitals,[7,22,41,107] and our results confirm this point. To clearly elucidate the influence of solvent effects, we performed the CM5 charge analyses for representative molecular dynamics snapshots. The comparison of the results in vacuum and in water reveals that water molecules alone do not significantly enhance electron back-donation to the adsorbed N_2 (−0.31 e in vacuum to −0.33 e in water). The presence of protons in water induces

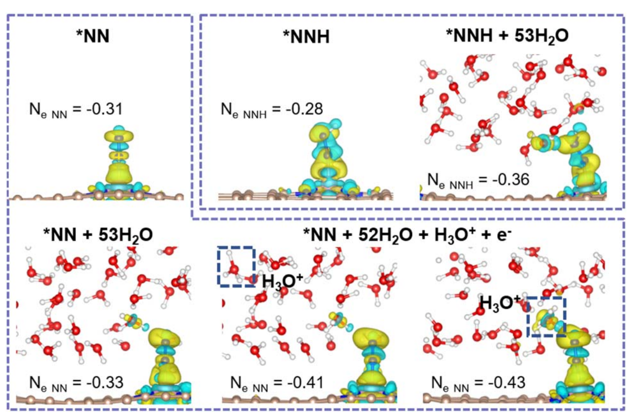

Fig. 6 Charge density difference (denoted by $\Delta\rho$) isosurfaces of absorbed species on the Ru–N$_4$–C catalyst. For vacuum systems, $\Delta\rho = \rho_{total} - \rho_{*} - \rho_{N_2 or N_2H}$, and for systems in water, $\Delta\rho = \rho_{total} - \rho_{*+water} - \rho_{N_2 or N_2H}$, where the subscript "total" of ρ denotes *NN or *NNH in vacuum or in water, and the subscript "*" denotes the Ru–N$_4$–C catalyst. Yellow and cyan regions represent electron accumulation and depletion, respectively, with an isosurface value of 0.003 e bohr^{-3}. The CM5 charges located in the NN or NNH fragment of *NN or *NNH are also given.

a strong local electric field that significantly increases the negative charge accumulation on N_2 (peaking at -0.43 e in close proximity), thereby strengthening the activation of the N–N bond. This finding suggests a synergistic coupling between the local electric field of the approaching ion and the electron back-donation mechanism. The proximity of ions is vital for the proton transfer process, highlighting the potential benefits of engineering proton-delivery channels or manipulating interfacial cations in future catalyst design.[56,107]

4 Conclusion

In this work, we performed long-time deep potential molecular dynamics simulations with enhanced sampling to investigate the electrocatalytic NRR process by the Ru–N_4–C single-atom catalysts in water, in particular focusing on the explicit solvent effect on the competitive adsorption of nitrogen molecules and protons at the reactive site and on the first protonation process of NRR. Our study provides detailed dynamic processes and directly reveals the significant effect of the inclusion of explicit solvent water molecules on the quantitative description of these key processes in the electrocatalytic NRR. The adsorption free energies of the nitrogen and the proton on the Ru–N_4–C in water are determined as -0.7 eV and -0.3 eV based on the DPMD simulations, respectively, which are smaller in absolute values than those calculated by the CHE model (-1.1 eV and -0.8 eV) without explicit water. It shows that the solvent effect decreases the adsorption intensity of the Ru–N_4–C catalyst toward both the nitrogen and proton to some extent, but does not change the dominance of nitrogen adsorption in thermodynamics. At the same time, the DPMD simulations provide a prediction of the free energy barrier for the dynamic processes of the nitrogen and proton adsorption on the catalyst in water, which is 0.3 eV for both.

For the rate-determining step of NRR on the Ru–N_4–C catalyst, namely the hydrogenation step of adsorbed N_2, we reveal that both the hydronium ion and water molecule near the adsorbed N_2 can participate in the reaction as reactants, but in the present acidic aqueous solution, the pathway with the hydronium ion as a source of protons is more favorable in dynamics. Furthermore, this reaction is driven by the synergistic effect of the solvation shell and the strong local electric field induced by the approach of the proton donor. The present explicit-solvent DPMD simulations successfully capture the energetic gains from the interaction with the explicit solvents which cannot be described by the static calculations with the CHE model in a vacuum. Consequently, we find that the explicit solvent environment enhances the thermodynamics and kinetics of the first hydrogenation step: the reaction free energy is reduced to only 0.15 eV, much lower than the CHE result of 1.19 eV, and the free energy barrier is predicted to be 0.59 eV, which is in good agreement with experimental observations.

Conflicts of interest

The authors declare no competing financial interest.

Data availability

The simulation and visualization software packages employed for performing the calculations are freely accessible. The necessary input files for replicating the research can be obtained at https://github.com/Zhang-pchao/research/tree/main/OPES-DPMD-NRR.

The data that support the findings of this study are available in the supplementary information (SI) of this article. Supplementary information: DP model training, dataset building, enhanced sampling, and other computational details. See DOI: https://doi.org/10.1039/d5ta09029f.

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